PortGuard AI — SteelHacks XIII

12-slide presentation script

September 20, 2026

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SteelHacks XIII — 12-slide presentation script

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Sources

PortGuard AI

SteelHacks XIII — 12-slide presentation script

Tracks: Xtract (Signal-to-Insight Engine) · Beyond the Chatbot (Best Use of NVIDIA Nemotron)

Live: pixelsplease.vercel.app · **API:** portguard-api.onrender.com

Every figure in this deck is either externally cited or measured from the running system. Nothing is estimated for effect.

Slide 1 — Title

Visual: full dashboard screenshot — Risk map tab, carrier index open, 33 vessels on the map.

PortGuard AI

MARITIME DISRUPTION → SHIPMENT-SPECIFIC, CONTRACT-AWARE DECISIONS

25 live sources · 926 signals · 60 clustered events · 33 vessels

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Slide 2 — Problem statement

Visual: the Traffic tab chart, or a three-row table of the chokepoint drops.

A disruption is public within minutes. Its cost to *your* container takes days to work out.

When Hormuz closes, three questions decide the money:

1. How long if we wait?
2. Can we legally reroute — does *this* Bill of Lading permit deviation?
3. Does *this* force majeure clause cover *this* event, and when does notice expire?

Today that is a person reading news, then a PDF contract, then a calculator. Per container. **The notice window often expires before the answer arrives.**

Slide 3 — Who actually loses the money

Visual: two-column contrast — “Carriers” against “Cargo owners”.

The counter-intuitive part: carriers profited. Cargo owners paid.

Measure	Figure
Maersk 2024 EBIT	**$6.5B, +65% year on year** (rates rose with the crisis)
Maersk 2025 guidance	**$0 EBIT if the Red Sea reopens**, **$3.0B if rerouting continues**
Demurrage and detention, 2024	**more than $10B** added to global ocean freight costs (Drewry)
US FMC finding, 9 carriers, Apr 2020 – Mar 2023	**$12.9B** collected in D&D
Cape of Good Hope reroute	**+$200–400 per TEU**

The loss sits with the **beneficial cargo owner** — the manufacturer, retailer or importer. They pay demurrage, detention and late-delivery penalties on contracts they did not get to reprice.

> **Say this out loud.** Everyone assumes the shipping lines were the victims of the Red Sea crisis. They were the beneficiaries. That reframing is the pitch.

Slide 4 — How we identified the problem

Visual: the All signals tab — article cards from mixed sources.

We started from the data, not from an idea.

1. Pulled IMF PortWatch daily transit counts for 2,065 ports and every major chokepoint.
2. Measured the drops ourselves against a 90-day baseline. **These are not taken from a press release** — they are computed in the product:

Chokepoint	vs 90-day baseline	Transits per day
Kerch Strait	**–100.0%**	0.0 against 3.6
Strait of Hormuz	**–42.8%**	5.3 against 9.2
Bosporus Strait	**–25.3%**	52.3 against 70.0
Bab el-Mandeb Strait	**–13.8%**	27.3 against 31.7
Suez Canal	**–8.0%**	37.9 against 41.2

As of 2026-09-13. Source: IMF PortWatch daily traffic.

3. The disruption data was **free and public**. The contract exposure was the part nobody had connected to it.

Slide 5 — Databases we searched

Visual: the sidebar source list — 25 sources with their fetch times.

25 sources wired. We also documented what does not work.

Working, and live in the product:

- IMF PortWatch — 2,065 ports, 4,897 daily traffic rows
- GDACS hazard alerts
- **NGA NAVAREA** broadcast warnings — JSON, 386 in force
- **NOAA National Hurricane Center** — three basins
- **US Navy Joint Typhoon Warning Center**
- **USGS** — M4.5+ earthquakes, proximity-matched to ports
- 17 maritime trade-press RSS feeds — JOC, The Loadstar, gCaptain, Maritime Executive, Seatrade, Splash247, Container News, World Cargo News, Marine Link, Maritime Logistics, Marine Insight, Bunkerspot, Hellenic Shipping News, SeaNews, FreightWaves, Supply Chain Dive, Offshore Energy
- GDELT and Google News

Rejected, with reasons:

- **UKMTO and MARAD MSCI return 403 to any server.** The two most authoritative sources for Hormuz and the Red Sea, blocked behind a WAF.
- Lloyd's List and TradeWinds — paywalled, 500 and 404 without a subscription.
- **AIS is a paid product.** So we do not claim live vessel tracking anywhere.

> Judges respect knowing what you rejected and why. This slide is credibility.

Slide 6 — Existing solutions and the gap

Visual: three columns — Track → Alert → **Decide**, with the last one highlighted.

Visibility platforms stop one step short.

Platform	What it answers
project44, FourKites	*Where is my container?*
Windward, Everstream	*Is there a disruption?*
Portcast, Sinay	*What is the new ETA?*

All three answer questions about **the world**. None answers a question about **your contract**.

The gap we address: nobody prices *wait against reroute against force majeure* using **your** Bill of Lading's demurrage rate, deviation clause and notice deadline — and then drafts the notice.

Disruption data is commoditised. **The contract is where the money is, and it is where nobody is looking.**

Slide 7 — Real data against simulated

Visual: an expanded shipment row showing the provenance line.

We label every row. Provenance is a database column, not a footnote.

Real and verifiable:

- 926 signals across 25 sources, every one with a live source URL
- 2,065 ports and 4,897 daily traffic rows from IMF PortWatch
- **33 real container vessels across 11 real carriers** — Maersk, MSC, CMA CGM, COSCO, Evergreen, OOCL, HMM, NYK, ONE, ZIM, APL
- **Every IMO number validated against its own check digit.** The validator caught two bad numbers during the build. Look any vessel up in Equasis or GISIS.

Simulated, and flagged in the schema as `terms_source = 'synthetic'`:

- Contract terms, BL numbers, shipper and consignee names, cargo values
- *Why:* real Bills of Lading are commercially confidential and published nowhere. The Xtract rules explicitly permit synthetic documents you create yourself.

Estimated, and stated in the interface:

- Vessel positions — computed from elapsed ETD to ETA along a real sea route. **Not AIS. The map says so.**

Slide 8 — Solution

Visual: pipeline diagram — Ingest → Cluster → Match → Price → Draft → Approve.

Six stages, from a headline to a signed notice.

1. **Ingest** — 25 sources, each on its own schedule
2. **Cluster** — reports become events; confidence weighted by *who* reported it, not only how many did
3. **Match** — events to shipments by port and chokepoint
4. **Price** — wait, reroute or force majeure, from that Bill of Lading's own terms
5. **Draft** — force majeure notice and customer email
6. **Approve** — **nothing sends without a human.** 30 drafts waiting, 0 sent automatically

Credibility tiering is what makes extra sources useful rather than noisy:

```
confidence = tier_floor + 0.15 × (publishers − 1)
```

Tier 1 sources — NOAA, NGA, IMF PortWatch — floor at 0.75. Tier 4 aggregators floor at 0.35. **One NOAA advisory outranks three aggregators repeating each other.**

Slide 9 — The calculation

Visual: the "What each option costs" panel — Monitor $20,000, Reroute $48,000, Force majeure $0 recommended.

Rule-based, auditable, and no model guessing at money.

Wait = `delay_days × late_delivery_penalty` plus demurrage and detention past free time. Delay comes from a table by event type and severity — a chokepoint disruption is **2, 6 or 12 days** for low, medium and high severity.

Reroute = `extra_days × penalty` plus a `$1,500 per container` surcharge. Real alternatives at real distances:

Chokepoint	Alternative	Extra days
Suez Canal, Bab el-Mandeb	Cape of Good Hope	+12
Panama Canal	Suez or Cape Horn	+10
Malacca Strait	Lombok and Makassar	+3
Dover Strait	North of Scotland	+2
Strait of Hormuz, Bosporus	**None**	**no sea route avoids them**

The model returns `None` rather than inventing an alternative that does not exist.

Force majeure = demurrage and detention only. **It excuses the late-delivery penalty, not the port charges.** Available only when the clause names that event type and notice falls inside the contractual window.

> Worked example on screen: Monitor $20,000 · Reroute $48,000 · **Force majeure $0, recommended.** Notice due 2026-09-21 19:32 UTC.

Slide 10 — Nemotron: Beyond the Chatbot

Visual: architecture diagram with the Nemotron stages highlighted.

Nemotron is in the pipeline, not bolted onto a chat box.

Stage	What Nemotron does
`xtract.py`	Refines raw signals — event type, severity, location — from unstructured news
`run_nemotron.py`	**The decision engine.** Structured-JSON risk, action and rationale per event and shipment
`ask.py`	Grounded question answering, cited, over the live database
`eval_model.py`	Our own benchmark harness

Evidence it works, including a failure we found:

- Measured `reasoning_effort=none` as **3 to 4 times faster and more stable** on nemotron-3-nano:4b. Found by benchmarking, not by guessing.

- **The failure:** the hosted model ran at **about 39 seconds** on analytical questions. The documented 20-second budget **never fired**, because `requests` applies its timeout as a per-read gap rather than a total deadline — a model trickling tokens resets that clock indefinitely.
- **The fix:** a deterministic router answers the 194 most common questions from SQL in **about 1 millisecond**, and Nemotron keeps the genuinely open-ended ones. Roughly **39,000× faster on the hot path**, with the model where it earns its keep.

Slide 11 — Xtract fit

Visual: a decision panel showing numbered, clickable citations.

Against the four criteria, directly.

"Support for different sources without rebuilding everything." Fetchers are pluggable. **We added 16 sources in a single session**, taking it from 9 to 25. A `STRICT` flag filters low-purity feeds — FreightWaves measured 23 of 56 items as maritime.

"A model that can find useful signals in incoming documents." Nemotron Xtract over news text, alongside a deterministic classifier. 926 signals resolve into 60 events.

"Insights that link back to where they came from." Every signal carries its `source_url`. Every event carries report and publisher counts. Every decision carries a citation list, rendered as clickable links. *HORMUZ-26: 108 reports from 83 publishers, confidence 0.99. Click through to any one of them.*

"A clean way to actually see and use the output." A live dashboard on one screen, with no page scrolling.

Rules: GDELT, RSS feeds and press releases — yes. Synthetic documents we created ourselves — yes. **No LANXESS data** — correct, none is used.

Slide 12 — Demo

Visual: the live application. Do not use slides for this section.

Ninety seconds, in this order:

1. **Sidebar** — 25 sources, all fetched minutes ago. It is live right now.

2. **Risk map** — 33 real vessels. Click **COSCO Shipping Lines** in the carrier index; its three ships light up and the other thirty fade. Hover one for a real IMO number, a real carrier, and an honest “estimated, not AIS”.
3. **Shipments** — click **MAERSK SALALAH**. The detail opens underneath it: HIGH risk, Red Sea, **notice due in 33h 21m**.
4. **The calculation** — Monitor $20,000, Reroute $48,000, **Force majeure $0**. The cheapest option, derived from that Bill of Lading’s own clause.
5. **The draft** — a force majeure notice, already written, awaiting approval. **Not sent.**
6. **Ask PortGuard** — *”How many ships does Cosco Africa have in transit?”* Answers instantly, and corrects the premise: COSCO AFRICA is a ship, not a company.

Close on: *”Everything you have just seen runs on 25 live public sources. The only synthetic part is the contract terms — and the database labels every one of them.”*

Two things to fix before presenting

1. **The `$0` force-majeure figures look like a bug to a judge.** They are correct — force majeure waives the late-delivery penalty, and these Bills of Lading still have free time left, so there is no demurrage yet. Say that out loud or the audience will assume the number is broken.
2. **Check the cron is firing** so the sidebar reads green rather than amber on stage. An amber “the ingest pipeline may not be running” warning during a live demo undercuts the whole freshness argument.

Sources

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